Final Examination

Prof. J. Whitfield

Thursday, August 8, 2024 Time allowed: 3 hours Total: 100 points

Instructions. This exam has two long problems of 50 points each, each built around a case. Answer all parts. Point values are in brackets; budget roughly 1.7 minutes per point. Explain your reasoning; complete strategies must specify an action at every decision point, including those not reached in equilibrium. You may use a calculator and one sheet (both sides) of your own notes. Fractions or decimals (to three places) are both acceptable.

Name: _____ Student ID: _____

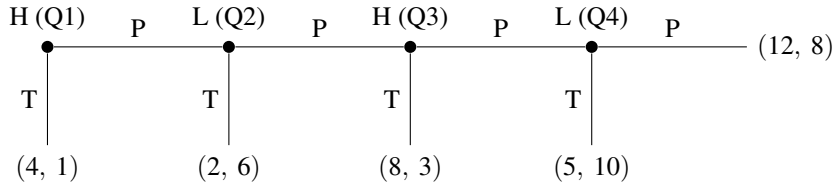
	1, Part I	1, Part II	2	Total
Points	20	30	50	100
Score				

Problem 1: The Harbor ferry venture [50 points]

The Harbor Authority (H) and Coastal Line (L), a ferry operator, run a joint venture on a new ferry service. The venture's value grows every quarter it survives, but either partner may unilaterally dissolve it when it is her turn to review the contract, seizing the larger share of the assets.

Part I: Growing the venture [20 points]

Reviews alternate: H reviews in quarters 1 and 3, L in quarters 2 and 4. At each review the reviewing partner either *Takes* (dissolves the venture) or *Passes*. If both partners pass at every review, the venture is sold at the end of quarter 4. Payoffs (H, L) in \$ millions are:



(a) Find the subgame perfect equilibrium by backward induction. What is the equilibrium outcome, and how much of the potential value of the venture is lost? [5]

(b) How many pure strategies does each partner have? List H's pure strategies. [4]

(c) Prove that in *every* Nash equilibrium, pure or mixed, H takes at the first review with probability one. (*Hint:* consider the last review that is reached with positive probability.) [5]

(d) Before the venture starts, H can sign a binding contract promising to pay L a bonus of $b \geq 0$ if the venture survives to the sale at the end of quarter 4, so the final payoffs become $(12-b,\ 8+b)$. Assume

a partner who is indifferent between Take and Pass chooses Pass. For which values of b does the SPE have both partners passing at every review? Which b would H choose, and what are the resulting payoffs? Explain what happens if H promises $b = 4.5$. [6]

Part II: Negotiating the terminal lease [30 points]

The partners (who, as you may have guessed, dissolved the venture) must still agree on how to split the value of a long-term terminal lease that they hold jointly. Normalize its value to 1. They bargain by alternating offers: in round 1 one partner proposes a split, the other accepts or rejects; after a rejection, the other partner makes the next proposal, and so on. Each round of delay multiplies a partner's payoff by her discount factor: $\delta_H = 0.9$ for the Harbor Authority and $\delta_L = 0.8$ for Coastal Line. A responder who is indifferent accepts. Unless stated otherwise, H proposes in round 1.

(e) Suppose there is only one round: if L rejects H's offer, both get 0. Find the SPE. [4]

(f) Suppose there are at most three rounds (H proposes in rounds 1 and 3, L in round 2; if the round-3 offer is rejected, both get 0). Find the SPE offers and the agreed split. [6]

(g) Now suppose there is no deadline. Find the SPE shares using the stationarity conditions of the Rubinstein model. State the equilibrium strategies completely (what each partner proposes and what she accepts, in every round) and describe the equilibrium outcome. [8]

(h) Suppose instead that L proposes first in the infinite-horizon game. What are the shares now? Use your answers to (g) and (h) to comment on the roles of patience and of moving first. [6]

(i) Let rounds last Δ months, with $\delta_H = e^{-r_H \Delta}$ and $\delta_L = e^{-r_L \Delta}$, where $r_H = 0.04$ and $r_L = 0.12$ per month. Find the limit of H's infinite-horizon share (with H proposing first) as $\Delta \to 0$, and interpret the result. [6]

Problem 2: The memory-chip market [50 points]

Arcadia Semiconductors (A) and Brightwave (B) are the only producers of a specialized memory chip. Inverse demand is $P = 100 - Q$, where $Q = q_A + q_B$. Arcadia's constant marginal cost is $c_A = 10$ and Brightwave's is $c_B = 16$; there are no fixed costs.

(a) The firms choose quantities simultaneously (Cournot). Derive both best-response functions and find the Nash equilibrium quantities, price and profits. [8]

(b) Show that the equilibrium in (a) is the unique rationalizable outcome by iterated elimination of never-best responses. Specifically, report the intervals of quantities that survive rounds 1, 2 and 3 for each firm, and argue that the intervals shrink to the equilibrium. [8]

(c) Suppose Arcadia chooses its quantity first and Brightwave observes it before choosing (Stackelberg, with Arcadia as leader). Find the SPE quantities, price and profits. Who gains and who loses relative to Cournot? [8]

(d) Suppose instead the firms compete in prices (Bertrand), selling identical chips. Consumers buy from the cheaper firm; if prices are equal, all consumers buy from Arcadia. Assume that neither firm ever

sets a price below its own marginal cost. Find the Nash equilibrium prices and profits, and compare Arcadia's profit and consumer surplus with those under Cournot. [8]

(e) Brightwave licenses Arcadia's process, so from now on both firms have marginal cost 10. The firms interact every quarter forever, choosing quantities, and discount at factor δ per quarter. They consider splitting the monopoly output equally, supported by grim-trigger strategies that revert to the Cournot equilibrium forever after any deviation. Find the collusive, deviation and punishment profits and the smallest δ that sustains collusion. If $\delta = e^{-r/4}$ for a continuously compounded annual interest rate r, what is the largest r for which collusion is sustainable? [10]

(f) Suppose $\delta = \tfrac{1}{2}$, so full collusion is not sustainable. The firms instead agree that each produces q, with $22.5 \le q \le 30$, again with Cournot reversion after any deviation. Show that the smallest discount factor sustaining this agreement is $\delta(q) = \dfrac{9(30-q)}{150-q}$. Evaluate it at $q = 25$, and find the most profitable symmetric agreement the firms can sustain when $\delta = \tfrac{1}{2}$. [8]

— End of examination —